Single-Cell Data Analysis Workshop – October 2026

Single Cell Data Analysis Workshop is organized by Bioinformatics and Biostatistics Core (BBC), HPC Teams, and Genomics Cores.

Session I: Tuesday, Oct 20, 9:30 AM -11:30 AM – HPC refresher, single cell experiment design, and basic concepts.
Location: 2101 & 2102

Session II: Wednesday, Oct 21, 9:30 AM -11:30 AM – Single cell QC, filtering, normalization, dimensional reduction, and clustering.
Location: 3104 & 3105

Session III: Thursday, Oct 22, 9:30 AM -11:30 AM – Marker gene identification, annotation, and DE gene expression
Location: 3104 & 3105

Only participants who register for a session will receive a calendar invitation for that session

Beverage and light refreshments will be provided.
Space is limited. You must bring a laptop to attend the workshop.

Link to the registration form:

Single-Cell Data Analysis Workshop – October 2026 – Fill out form

Workshop overview

Day 1	HPC and OnDemand refresher, sc library prep, RNA sequencing and single cell concepts	Create an unfiltered Seurat object with counts and metadata
Day	**Main focus**	**Participant outcome**
Day 2	Quality control, normalization, dimensionality reduction, and clustering	Produce a filtered object with PCA, UMAP, and clusters
Day 3	Cell-type annotation, differential expression, interpretation, and data reproducibility.	Generate an annotated UMAP, marker table, and example comparison

Each day will be 2 – 2.5 hours

Day 1

Time	Session	presenter
5-10 min	Welcome, workshop goals, and dataset overview	Scott Givan
30 min	HPC and OnDemand refresher	Kaitlyn Denhaan (HPC)
20 min	Single-cell library preparation and sequencing	Genomics Core
10 min	Break	
45 – 50 min	Basic RNA sequencing and single-cell concepts	Ian Beddows

Day 2

Time	Session	presenter
10 min	Day 1 review and questions	Bobby Fillinger
50 min	Quality-control and filtering	
10 min	Break	
55 mins	Normalization, PCA, clustering, and UMAP	Kin Lau
5 min	Summary and checkpoint	

Day 3

Time	Session	presenter
10 min	Day 2 review and questions	Daisy Fu
50 min	Marker genes and cell-type annotation	
10 min	Break	
45 min	Differential-expression and experimental-design concepts	Daniel Bautista
15 min	Interpretation, exporting results, and reproducibility	
5 min	Final questions	